# Refined Use Case Descriptions

## UC-01 - Request Carpark Recommendation

| | |
|---|---|
| **Use Case ID** | UC-01 |
| **Use Case Name** | Request Carpark Recommendation |
| **Created By** | Nestor |
| **Last Updated By** | Nestor |
| **Date Created** | 30 August 2026 |
| **Date Last Updated** | 30 August 2026 |
| **Actor** | Anonymous Driver; Parking Data Provider; Destination, Map and Routing Provider |
| **Description** | The driver requests ranked carpark recommendations near a destination. ParkChoiceSG applies one fixed Best Overall scoring model using current parking availability, estimated parking cost, and walking distance, then explains why each result is recommended. |
| **Preconditions** | 1. ParkChoiceSG is running.<br>2. The driver can access the carpark recommendation interface. |
| **Postconditions** | 1. On success, the driver receives an explained, time-labelled list of up to five recommendations.<br>2. On failure, the driver receives a clear message that no reliable recommendation can be produced.<br>3. No account or profile is created. |
| **Priority** | Must Have |
| **Frequency of Use** | High - normally once per parking search or journey. |

| | |
|---|---|
| **Flow of Events** | 1. The driver opens the carpark recommendation function.<br>2. The driver enters a destination and intended parking duration, then submits the request.<br>3. ParkChoiceSG validates the inputs and resolves the destination to a location.<br>4. ParkChoiceSG identifies supported nearby carparks and obtains their latest reported availability, published parking rates, and walking-route measurements to the destination.<br>5. ParkChoiceSG verifies that each candidate has sufficiently current and comparable evidence. It labels every retained availability value as "Current reported availability" and shows its source timestamp.<br>6. ParkChoiceSG estimates the parking cost for the entered duration, calculates each candidate's fixed Best Overall score using current availability, estimated cost, and walking distance, and ranks the candidates. 7. ParkChoiceSG presents the highest-ranked carpark as the Best Overall recommendation, followed by up to four alternatives. Each result shows its raw factor values, timestamp, relevant warnings, and a concise explanation of its ranking. |
| **Alternative Flows** | UC-01.AC.1 - Invalid destination or duration<br>1. At Step 3, ParkChoiceSG identifies an invalid or unsupported input.<br>2. ParkChoiceSG explains which input must be corrected.<br>3. The use case resumes at Step 2.<br>UC-01.AC.2 - Destination cannot be resolved<br>1. At Step 3, the destination provider cannot resolve the entered destination<br>unambiguously.<br>2. ParkChoiceSG asks the driver to refine the destination or select a suggested match.<br>3. The use case resumes at Step 2 or Step 3 after the driver responds.<br>UC-01.AC.3 - Some external data is unavailable<br>1. At Step 4, one provider or part of its data is unavailable.<br>2. ParkChoiceSG continues only with carparks that still have the current availability, verified rate information, and walking-route measurement required for the fixed comparison.<br>3. ParkChoiceSG displays a warning that some carparks could not be assessed because data was unavailable.<br>4. The use case resumes at Step 5.<br>UC-01.AC.4 - A carpark has missing, stale, or incomparable evidence<br>1. At Step 5, ParkChoiceSG detects that a carpark lacks the minimum evidence required for a fair comparison.<br>2. ParkChoiceSG excludes that carpark instead of inventing or silently substituting a value.<br>3. The use case resumes at Step 5 with the remaining candidates.<br>UC-01.AC.5 - Fewer than five suitable carparks are available<br>1. At Step 7, fewer than five carparks satisfy the evidence and eligibility rules.<br>2. ParkChoiceSG displays every suitable result and states that no additional<br>comparable carparks were found.<br>UC-01.AC.6 - No suitable carpark can be recommended<br>1. At Step 5, no candidate has enough current and comparable evidence for<br>the fixed scoring model.<br>2. ParkChoiceSG explains why a reliable ranking cannot be produced and suggests changing the destination or duration, or retrying later.<br>3. The use case ends without a recommendation. |

| | |
|---|---|
| | UC-01.AC.7 - Driver selects a result<br>1. After Step 7, the driver selects a recommended carpark.<br>2. ParkChoiceSG starts UC-02 for the selected carpark. |
| **Exceptions** | UC-01.EX.1 - All required data sources are unavailable<br>1. ParkChoiceSG cannot obtain enough current evidence to produce a responsible recommendation.<br>2. ParkChoiceSG informs the driver that recommendations are temporarily unavailable and invites the driver to retry later.<br>3. The use case ends without a recommendation.<br>UC-01.EX.2 - Unexpected system error<br>1. ParkChoiceSG records the error without exposing credentials or sensitive technical details.<br>2. ParkChoiceSG informs the driver that the request could not be completed and provides a retry option.<br>3. The use case ends. |

| | |
|---|---|
| **Includes** | None. |
| **Special Requirements** | 1. When every required external dependency responds within one second, at least 95% of recommendation requests should be displayed within three seconds.<br>2. Every availability value must be labelled "Current reported availability" and accompanied by its source timestamp.<br>3. Every ranked result must show the availability, estimated cost, and walking distance used in its fixed Best Overall score, together with a concise reason for its position.<br>4. Missing or stale factor data must not be replaced with an invented value or presented as current.<br>5. Partial provider failure must be visible to the driver.<br>6. The core recommendation flow must not require an account, profile, origin, or device-location permission.<br>7. API credentials and provider tokens must not be embedded in the client application or exposed in user-facing errors.<br>8. A first-time driver should be able to submit a search using three required actions: enter a destination, enter a duration, and request recommendations. |

| | |
|---|---|
| **Assumptions** | 1. The driver is searching for an ordinary passenger-car parking space. 2. The driver uses ParkChoiceSG only while stationary or through a passenger.<br>3. Parking availability and rate data come from authorised sources and may change after retrieval.<br>4. Estimated cost is based on verified published rate information and the entered duration; it is not a payment quotation or guarantee. 5. Current availability does not guarantee that a lot will remain available when the driver arrives.<br>6. Turn-by-turn navigation is handed off to an external navigation service. 7. The scoring weights, normalisation rules, and tie-breaker are defined consistently in the project's scoring requirements and tests. |
| **Notes and Issues** | None. |

# UC-02 - Inspect Carpark Information

| | |
|---|---|
| **Use Case ID** | UC-02 |
| **Use Case Name** | Inspect Carpark Information |
| **Created By** | Nestor |
| **Last Updated By** | Nestor |
| **Date Created** | 30 August 2026 |
| **Date Last Updated** | 30 August 2026 |
| **Actor** | Anonymous Driver |
| **Description** | The driver selects a recommended carpark and inspects the information and evidence used to assess it. |
| **Preconditions** | 1. UC-01 has produced at least one carpark recommendation. |
| **Postconditions** | 1. The selected carpark's information is displayed.<br>2. The recommendation results remain unchanged.<br>3. No account or profile is created. |
| **Priority** | Must Have |
| **Frequency of Use** | High - normally after a recommendation request. |

| | |
|---|---|
| **Flow of Events** | 1. The driver selects a carpark from the recommendation results. 2. ParkChoiceSG opens the information view for the selected carpark. 3. ParkChoiceSG displays the carpark's name, address, current reported availability and timestamp, estimated cost for the entered duration, and walking distance to the destination.<br>4. ParkChoiceSG displays the carpark's Best Overall ranking explanation and any relevant data warnings.<br>5. ParkChoiceSG provides an option to open external navigation to the carpark. |
| **Alternative Flows** | UC-02.AC.1 - Selected carpark information is no longer available 1. At Step 2, ParkChoiceSG cannot retrieve the selected result from the active recommendation.<br>2. ParkChoiceSG explains that the information is no longer available. 3. ParkChoiceSG returns the driver to the current recommendation results.<br><br>UC-02.AC.2 - Some displayed information is no longer current 1. At Step 3, ParkChoiceSG detects that previously retrieved information no longer satisfies the freshness rule.<br>2. ParkChoiceSG marks the affected information as unavailable or outdated rather than presenting it as current.<br>3. ParkChoiceSG advises the driver to refresh the recommendation before relying on the result.<br>4. The use case resumes at Step 4 with the remaining information.<br><br>UC-02.AC.3 - Driver requests navigation<br>1. After Step 5, the driver selects the external-navigation option. 2. ParkChoiceSG starts UC-03. |

| | |
|---|---|
| **Exceptions** | UC-02.EX.1 - Unexpected system error<br>1. ParkChoiceSG records the error without exposing sensitive technical details.<br>2. ParkChoiceSG informs the driver that the carpark information could not be displayed.<br>3. ParkChoiceSG provides an option to return to the recommendation results.<br>4. The use case ends. |
| **Includes** | None. |
| **Special Requirements** | 1. Information displayed in UC-02 must use the same values, units, labels, and timestamps as the corresponding UC-01 result.<br>2. Availability must be labelled "Current reported availability" and must not be represented as a guaranteed parking space.<br>3. The ranking explanation must identify the availability, estimated cost, and walking distance considered by the fixed Best Overall score. 4. When the information is already available from the recommendation result, the information view should appear within one second for at least 95% of selections on a supported device.<br>5. Inspecting a carpark must not require an account, profile, origin, or device-location permission. |

| | |
|---|---|
| **Assumptions** | 1. The selected carpark belongs to the active recommendation result set. 2. Estimated cost and walking distance retain the parking duration and destination used in UC-01.<br>3. Parking information may change after it is retrieved. |
| **Notes and Issues** | None. |

# UC-03 - Open External Navigation

| | |
|---|---|
| **Use Case ID** | UC-03 |
| **Use Case Name** | Open External Navigation |
| **Created By** | Nestor |
| **Last Updated By** | Nestor |
| **Date Created** | 30 August 2026 |
| **Date Last Updated** | 30 August 2026 |
| **Actor** | Anonymous Driver; Destination, Map and Routing Provider; Device Operating System |
| **Description** | The driver requests directions to a selected carpark, and ParkChoiceSG hands the carpark destination to an external navigation service. |
| **Preconditions** | 1. UC-02 is displaying a selected carpark. |
| **Postconditions** | 1. On success, an external navigation service opens with the selected carpark as its destination.<br>2. On failure, the driver remains in ParkChoiceSG and receives a clear alternative. |
| **Priority** | Must Have |
| **Frequency of Use** | Medium - when a driver decides to travel to a recommended carpark. |
| **Flow of Events** | 1. The driver selects the option to navigate to the displayed carpark. 2. ParkChoiceSG obtains the selected carpark's verified name and destination coordinates.<br>3. ParkChoiceSG sends the destination to a supported external navigation service through the device operating system.<br>4. The external navigation service opens and displays a route to the selected carpark.<br>5. ParkChoiceSG completes the handoff without providing turn-by-turn navigation itself. |

| | |
|---|---|
| **Alternative Flows** | UC-03.AC.1 - More than one navigation service is available 1. At Step 3, the device operating system asks the driver to select a supported navigation service.<br>2. The driver selects a service.<br>3. The use case resumes at Step 4.<br>UC-03.AC.2 - No supported navigation application is available<br>1. At Step 3, the device cannot open a supported navigation application.<br>2. ParkChoiceSG keeps the carpark information visible and provides the<br>carpark name, address, and coordinates for manual use.<br>3. The use case ends without opening navigation.<br>UC-03.AC.3 - External navigation service cannot calculate a route<br>1. At Step 4, the external navigation service cannot produce a route.<br>2. The external service informs the driver according to its own failure<br>behaviour.<br>3. ParkChoiceSG remains responsible only for having supplied the correct<br>selected destination.<br>4. The use case ends. |

| | |
|---|---|
| **Exceptions** | UC-03.EX.1 - Selected destination is invalid or unavailable<br>1. At Step 2, ParkChoiceSG detects missing or invalid destination coordinates.<br>2. ParkChoiceSG does not launch the external navigation service. 3. ParkChoiceSG explains that navigation cannot be opened for this carpark and keeps its information visible.<br>4. The use case ends.<br><br>UC-03.EX.2 - Unexpected handoff error<br>1. ParkChoiceSG records the error without exposing sensitive technical details.<br>2. ParkChoiceSG informs the driver that navigation could not be opened and keeps the carpark information visible.<br>3. The use case ends. |
| **Includes** | None. |
| **Special Requirements** | 1. ParkChoiceSG must send only the selected carpark's name and destination coordinates required for the handoff.<br>2. ParkChoiceSG must not require or collect the driver's origin or device location for the handoff. The external navigation service manages any origin or location permission under its own controls.<br>3. The external application launch should begin within two seconds for at least 95% of requests on a supported device, excluding time spent inside the external service.<br>4. A handoff failure must not close the carpark information view or remove the selected result.<br>5. ParkChoiceSG must clearly indicate that navigation is provided by an external service. |

| | |
|---|---|
| **Assumptions** | 1. The device operating system supports external application links or intents.<br>2. At least one suitable external navigation service may be installed or accessible through a web browser.<br>3. ParkChoiceSG does not control route availability, traffic information, navigation accuracy, or permissions requested by the external service. |
| **Notes and Issues** | None. |

# UC-04 - Submit Error Report

| | |
|---|---|
| **Use Case ID** | UC-04 |
| **Use Case Name** | Submit Error Report |
| **Created By** | Kayen |
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | Anonymous Driver |
| **Description** | Allows the driver to anonymously flag data inaccuracies by reporting a carpark as having incorrect rates, being closed, or being full contrary to the displayed availability. The system submits each report for operator review without attaching any personal data. |
| **Preconditions** | 1. The driver has selected a recommended carpark and is viewing its information. |
| **Postconditions** | 1. The report is submitted to a queue for operator review, or queued locally if offline.<br>2. No personal data or account credentials are attached to the submission. |
| **Priority** | Low |
| **Frequency of Use** | As needed by the driver, up to the daily limit. |

| | |
|---|---|
| **Flow of Events** | 1. The driver selects the "Report incorrect information" option for a specific carpark.<br>2. The system presents the report categories - incorrect rates, closed, or full contrary to displayed availability - along with an optional description field.<br>3. The driver selects a category, optionally inputs a description, and submits the report.<br>4. The system verifies that the driver has not exceeded the limit of 5 reports per 24-hour period using a rotating per-installation token.<br>5. The system transmits the report to a queue for operator review without attaching any personal data.<br>6. The system confirms the successful submission to the driver. |
| **Alternative Flows** | UC-04.AC.1 - Daily limit reached:<br><br>1. At Step 4, the system detects that the driver has already submitted five reports today.<br>2. The system explains the daily limit to the driver and declines the submission. |

| | |
|---|---|
| **Exceptions** | UC-04.EX.1 - Report submission fails:<br><br>1. At Step 5, the system fails to transmit the report due to a lack of network connectivity.<br>2. The system queues the report locally on the device.<br>3. The system automatically retries the submission when connectivity is restored |
| **Includes** | None. |
| **Special Requirements** | 1. Reports must remain completely anonymous; no report may be traced to an individual.<br>2. The daily submission limit must be enforced using a rotating installation token that carries no user identity. |
| **Assumptions** | 1. The device operating system can securely generate and store the rotating per-installation token used to enforce the daily submission limit.<br>2. The driver submits the report based on actual, physical observation of the carpark's current status or rate boards.<br>3. The external parking data supplied to the system is occasionally flawed or outdated, making crowd-sourced error reporting a necessary fallback. |

| | |
|---|---|
| **Notes and Issues** | 1. Because the system strictly avoids holding user accounts and enforces complete anonymity, there is a vulnerability where malicious users could submit false reports up to their daily limit of 5.<br>2. The system must handle potential data inconsistencies or identifier mismatches between LTA feeds and HDB datasets when operators attempt to reconcile these user reports. |

# UC-05 - Authenticate Operator

| | |
|---|---|
| **Use Case ID** | UC-05 |
| **Use Case Name** | System Operator / Administrator |
| **Created By** | Kayen |
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | Anonymous Driver |
| **Description** | The operator provides administrative credentials over a secure channel to unlock administrative access privileges and render the hidden Operator Administrator console |
| **Preconditions** | 1. ParkChoiceSG application / portal is running and connected via TLS 1.2+.<br>2. The system administrative endpoint is reachable |
| **Postconditions** | 1. An encrypted bearer authentication token is issued and stored for the administrative session.<br>2. The hidden Operator console tab becomes visible and active, and grants access to system management tools |
| **Priority** | High |
| **Frequency of Use** | Daily (for operational checks and monitoring sessions) |

| | |
|---|---|
| **Flow of Events** | 1. Operator accesses the administrative login interface.<br>2. Operator enters their administrative username/email and administrative password.<br>3. System validates credentials against secure server-side configuration over TLS 1.2+.<br>4. System grants access, issues a session bearer token, and renders the Operator Administration dashboard (Report Queue and API Health overview). |
| **Alternative Flows** | *UC-05.AC.1 (Invalid Credentials)*: System notifies the operator of authentication failure and retains login view without detailing which field failed.<br><br>*UC-05.AC.2 (Session Timout)*: Operator remains inactive past maximum session limit. System invalidates the bearer token and redirects the operator back to the administrator login view |

| | |
|---|---|
| **Exceptions** | *UC-05.EX.1 (Connection Insecure/Offline)*: System aborts authentication attempt, shows a connection error message, and reports connection failure. |
| **Includes** | None. |
| **Special Requirements** | Authentication requests must process within 1 second, and credential input fields should enforce masked text entry |
| **Assumptions** | The operator has been provisioned valid administrative credentials prior to accessing the portal |
| **Notes and Issues** | Session tokens must be stored in secure, encrypted storage to prevent unauthorised administrative escalation |

# UC-06 - Manage User Error Reports

| | |
|---|---|
| **Use Case ID** | UC-06 |
| **Use Case Name** | Manage User Error Reports |
| **Created By** | Kayen |

| | |
|---|---|
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | System Operator |
| **Description** | Operator inspects anonymous data error reports filed by users (e.g., incorrect tariffs, unexpected lot closures, capacity mismatch) and marks them resolved or rejected. |
| **Preconditions** | 1. Operator has successfully completed UC-05 (<<include>>). |
| **Postconditions** | 1. The report status is updated with resolution notes, an audit timestamp, and operator identifier. |
| **Priority** | High |
| **Frequency of Use** | Continuous, on-demand during operator shift |
| **Flow of Events** | 1. Operator selects the "Report Queue" panel on the Operator tab.<br>2. System fetches and displays all active reports sorted in reverse chronological order.<br>3. Operator selects an individual report to review carpark details, reported category, timestamp, and user description.<br>4. Operator determines corrective action and selects "Mark Resolved" or "Reject".<br>5. Operator inputs mandatory administration resolution remarks.<br>6. System records outcome, closes the ticket, updates the queue display, and logs the administrative action. |
| **Alternative Flows** | UC-06.AC.1 - Carpark not in database<br>1. System identifies that reported carpark is not in database.<br>2. System automatically rejects report and archives it with an automated reason record.<br>UC-06.AC.2 - Apply Direct Override<br>1. Operator determines the report requires immediate system adjustment and transitions directly to UC-07 to apply a manual override on the flagged carpark. |
| **Exceptions** | UC-06.EX.1 - Empty Resolution Remarks<br>1. Operator attempts to submit a resolution without entering remarks.<br>2. System will block the submission and prompt for required administrative input. |
| **Includes** | UC-05 - Authenticate Operator |

| | |
|---|---|
| **Special Requirements** | 1. Report list updates in real time or through pull-to-refresh without page reload. |
| **Assumptions** | 1. User reports should contain no identifiable information per privacy standards. |
| **Notes and Issues** | Resolved reports should remain searchable in an archived library for administrative auditing. |

# UC-07 - Apply Manual Carpark Override

| | |
|---|---|
| **Use Case ID** | UC-07 |
| **Use Case Name** | Apply Manual Carpark Override |
| **Created By** | Kayen |
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | System Operator |
| **Description** | Operator manually modifies a carpark status (e.g. marked temporarily closed) and adjust rate properties to prevent erroneous data from reaching driver recommendations. |
| **Preconditions** | 1. Operator has authenticated via UC-05. |
| **Postconditions** | 1. Carpark record is updated with override flags and excluded from recommendation scoring if closed.<br>2. Overridden fields are locked against incoming daily automated source refreshes until manually withdrawn. |
| **Priority** | High |
| **Frequency of Use** | Needed when data discrepancies or physical carpark events occur |

| | |
|---|---|
| **Flow of Events** | 1. Operator searches for a specific carpark identifier or selects one from a flagged report.<br>2. Operator chooses an override action (e.g., "Mark Temporarily Closed", "Override Rate Table").<br>3. Operator supplies a mandatory administrative reason for the override.<br>4. System logs the override with timestamp, operator ID, and reason.<br>5. System excludes the carpark from all active recommendation calculations and displays closure banners on detail views. |
| **Alternative Flows** | UC-07.AC.1 - Withdraw Existing Override<br>1. Operator selects an active override and chooses "Withdraw Override".<br>2. System logs the removal and flags the carpark for source reconciliation during the next daily refresh. |
| **Exceptions** | UC-07.EX.1 - Data Store Write Lock<br>1. System fails to commit override settings due to database concurrency issues.<br>2. System will alert operator and maintain previous data state. |
| **Includes** | UC-05 - Authenticate Operator |
| **Special Requirements** | 1. Overrides must take effect across all recommendation queries upon submission. |
| **Assumptions** | 1. Manual overrides supersede all incoming API feed data until explicitly revoked by an operator. |
| **Notes and Issues** | Operator dashboard should feature an “Active Override” filter to prevent stale manual overrides from lingering indefinitely. |

# UC-08 - View API Health & Data Freshness

| | |
|---|---|
| **Use Case ID** | UC-08 |
| **Use Case Name** | View API Health & Data Freshness |
| **Created By** | Kayen |
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | System Operator, Parking Data Provider |
| **Description** | Operator monitors live telemetry for external data sources (LTA DataMall, Data.gov.sg, Google Maps, OneMap), reviewing uptime, request volumes, failure counts, and cache ages. |
| **Preconditions** | 1. Operator has authenticated via UC-05. |
| **Postconditions** | 1. Real-time integration metrics and endpoint telemetry are rendered without interrupting background polling services. |
| **Priority** | Medium |
| **Frequency of Use** | Continuous monitoring throughout operational shifts |
| **Flow of Events** | 1. Operator opens the "API Health" pane on the administrative dashboard.<br>2. System displays per-source metrics: request throughput, consecutive failure count, last successful sync timestamp, and cache age.<br>3. System highlights sources that exceed staleness thresholds (e.g., LTA poll older than 90s). |
| **Alternative Flows** | UC-08.AC.1 - Stale Source Manual Sync<br>1. Operator triggers UC-09: Trigger Manual Source Refresh directly from the health panel.<br>2. System sends immediate polling request to external provider and updates telemetry metrics. |

| | |
|---|---|
| **Exceptions** | UC-08.EX.1 - Telemetry stream Loss<br>1. System loses connection to internal telemetry logger.<br>2. Operator display flags telemetry as offline while background services continue operating. |
| **Includes** | UC-05 - Authenticate Operator |
| **Special Requirements** | 1. Telemetry board metrics must automatically refresh every 15 seconds without full page reloads. |
| **Assumptions** | 1. Data providers supply standard status headers or HTTP response codes during polling updates. |
| **Notes and Issues** | Automatic visual alerts such as state badges (green, yellow or red) should be used to make failing feeds immediately noticeable to the operator. |

# UC-09 - Trigger Manual Source Refresh

| | |
|---|---|
| **Use Case ID** | UC-09 |
| **Use Case Name** | Trigger Manual Source Refresh |
| **Created By** | Kayen |
| **Last Updated By** | Kayen |
| **Date Created** | 7 September 2026 |
| **Date Last Updated** | 7 September 2026 |
| **Actor** | System Operator / Administrator |
| **Description** | The operator triggers a manual refresh of a stale external data source to immediately update the system's cache. |
| **Preconditions** | 1. The operator has successfully authenticated via UC-05. |

| | |
|---|---|
| | 2. The operator is viewing the API Health panel on the administrative dashboard. |
| **Postconditions** | 1. The system updates the internal cache with fresh data from the external provider.<br>2. Existing manual carpark overrides are preserved against the fresh data. |
| **Priority** | Medium |
| **Frequency of Use** | On-demand when telemetry indicates that a data source has breached its freshness threshold. |
| **Flow of Events** | 1. The operator identifies a stale data source on the API Health pane and triggers a manual source refresh.<br>2. The system sends an immediate polling request directly to the external data provider.<br>3. The system receives the fresh data and updates the internal cache.<br>4. The system protects and preserves any existing manual operator overrides, ensuring they are not overwritten by the source data refresh.<br>5. The system updates the telemetry metrics on the API Health panel to reflect the newly successful sync timestamp and reset cache age. |
| **Alternative Flows** | None. |
| **Exceptions** | UC-09.EX.1 - External API Failure<br>1. At Step 2, the external API returns an error or malformed payload.<br>2. The system retries the failed request twice with an exponential backoff.<br>3. If the request still fails, the system retains the existing cached data, discards the payload, and increments the failure counter surfaced in the Operator health panel. |
| **Includes** | UC-05 - Authenticate Operator |
| **Special Requirements** | None. |

| | |
|---|---|
| **Assumptions** | 1. The external data providers (e.g., LTA DataMall, Data.gov.sg) support on-demand polling and supply standard HTTP response codes or status headers during the update.<br>2. The system backend has sufficient API quota remaining to execute the out-of-schedule refresh request without being blocked by the provider.<br>3. Manual operator overrides applied via UC-07 must remain active and override any conflicting data retrieved during this manual refresh. |
| **Notes and Issues** | 1. Frequent use of the manual refresh trigger could risk exceeding external API credit limits or free-tier constraints, which must be carefully managed due to the project's low budget.<br>2. Reconciling differing carpark identifiers between the newly fetched LTA DataMall availability feeds and the HDB/URA reference datasets remains an open technical risk during the refresh. |